



In the name of Allah

Application Kerman natural mineral biochar direction Dealing with frostbite

Kerman natural mineral biochar on biochemical and
physiological responses plants It is effective for frost stress



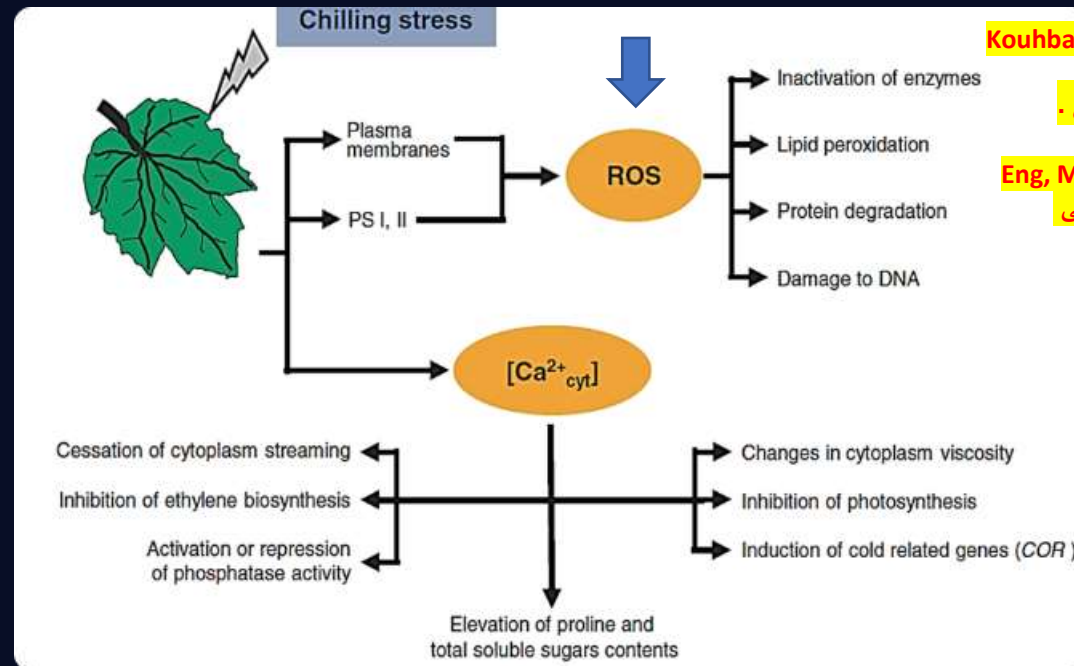
By Mr. Engineer Mohammad Majidi KouhBanani

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Effects of cold stress in plant cells



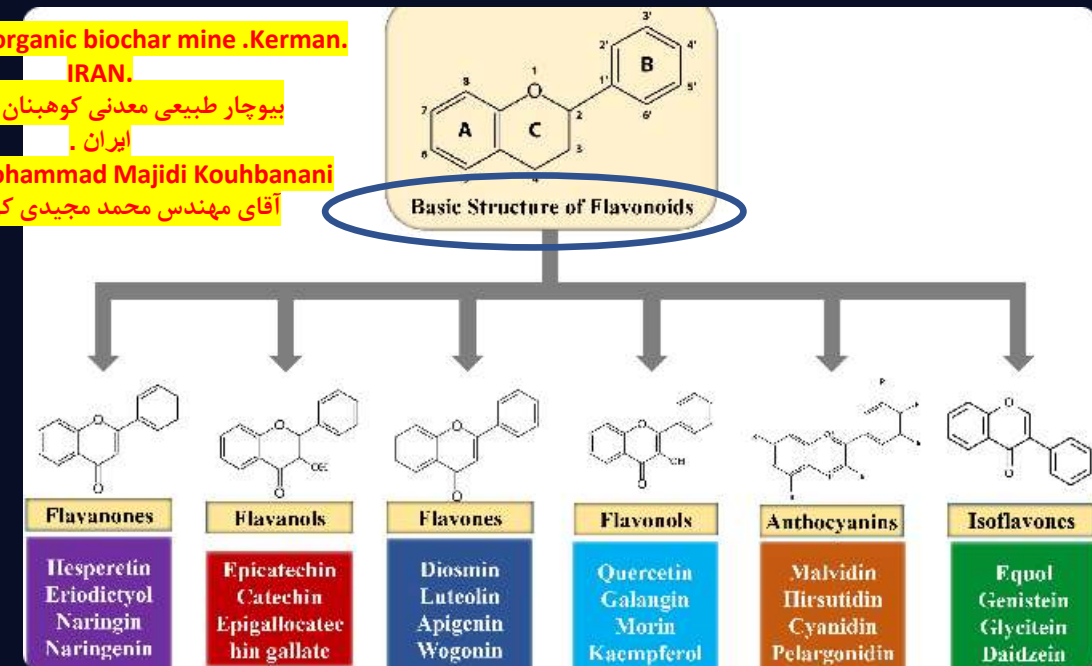
Accumulation of reactive oxygen species

This stress can increase the accumulation of reactive oxygen species and cause cell damage in plants

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The effect of increasing the amount of phenol and flavonoid in leaves

The amount of phenol and flavonoid in leaves can be effective in preventing frostbite and maintaining the health of leaves

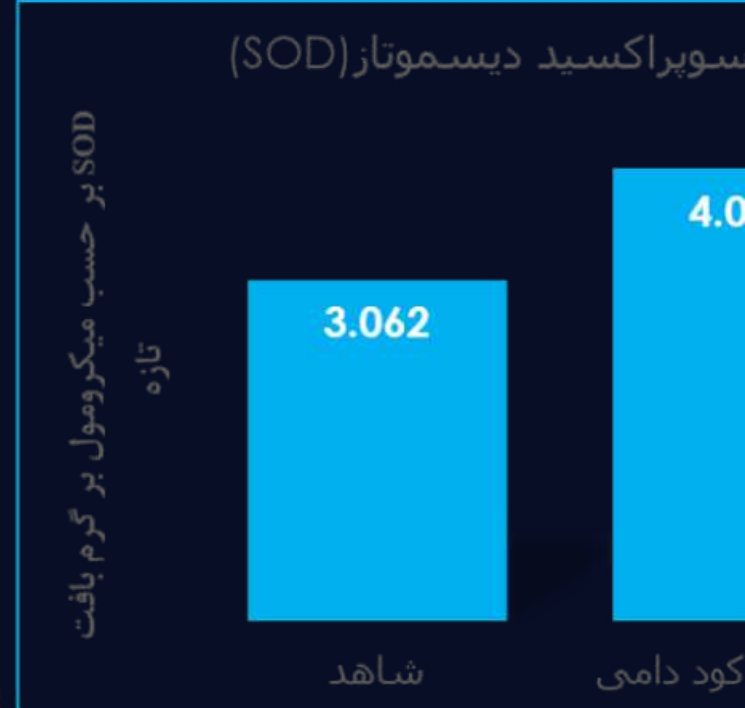
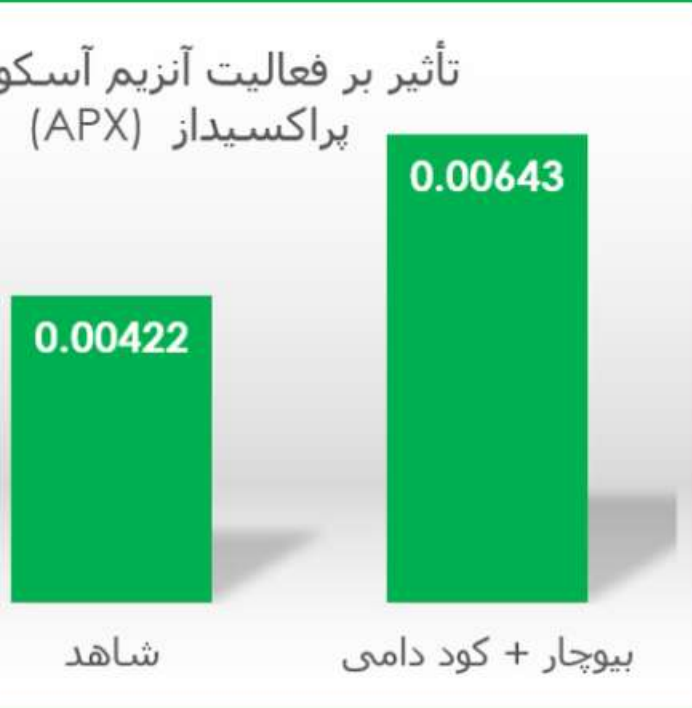
effects Kerman natural mineral biochar On Preventing the accumulation of reactive oxygen species

the effect Kerman natural mineral biochar On Activity of antioxidant enzymes

The effect of biochars on the growth of plants in stressful environments, including frost related to inhibiting the accumulation of reactive oxygen species, can be affected by increasing the activity of the antioxidant system.

the effect Kerman natural mineral biochar On Non-enzymatic antioxidants

By application Kerman natural mineral biochar The amount of non-enzymatic antioxidants has increased and this reduces the oxidative stress caused by frost stress.



Increasing the activity of ascorbate peroxidase and superoxide dismutase

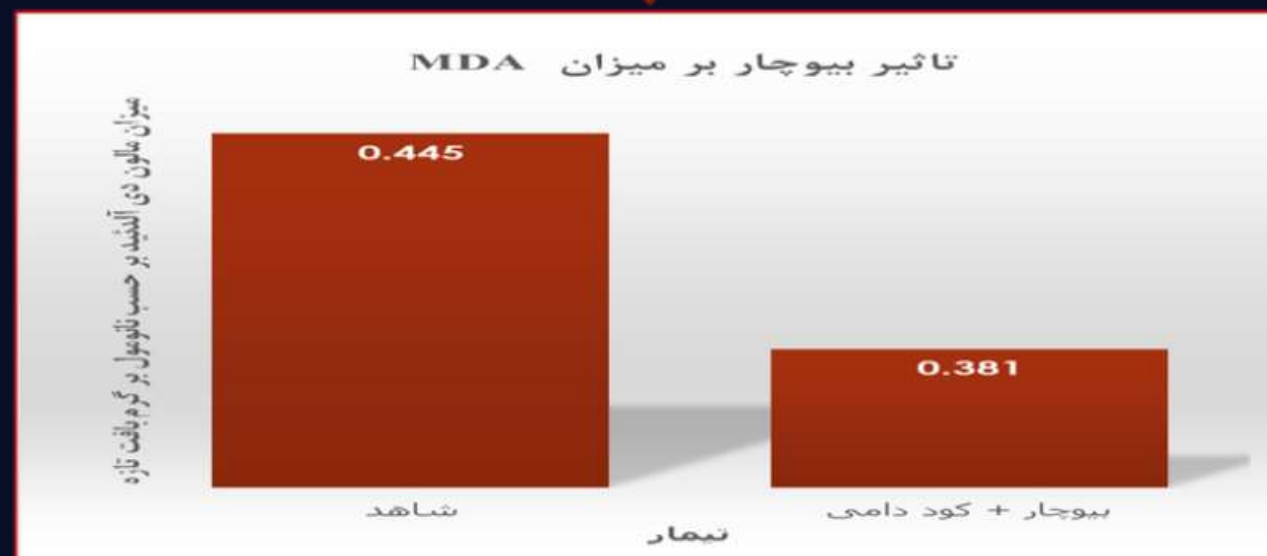
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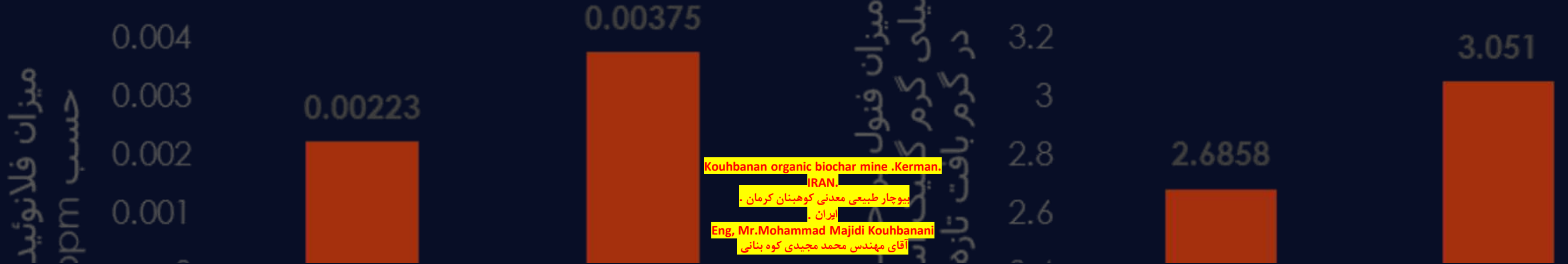
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مهار تجمع ROS

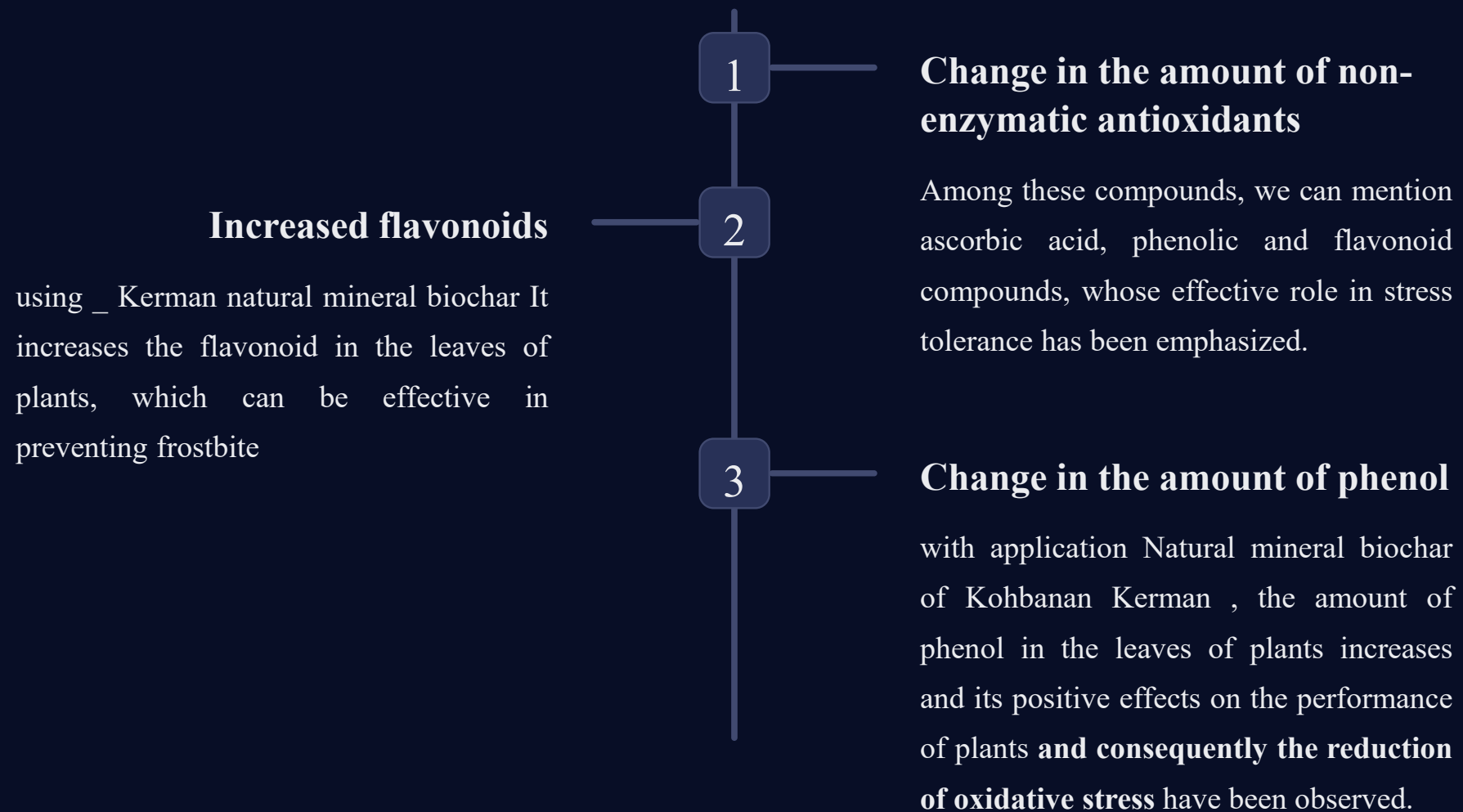


کاهش پراکسیداسیون لیپید و تجمع MDA

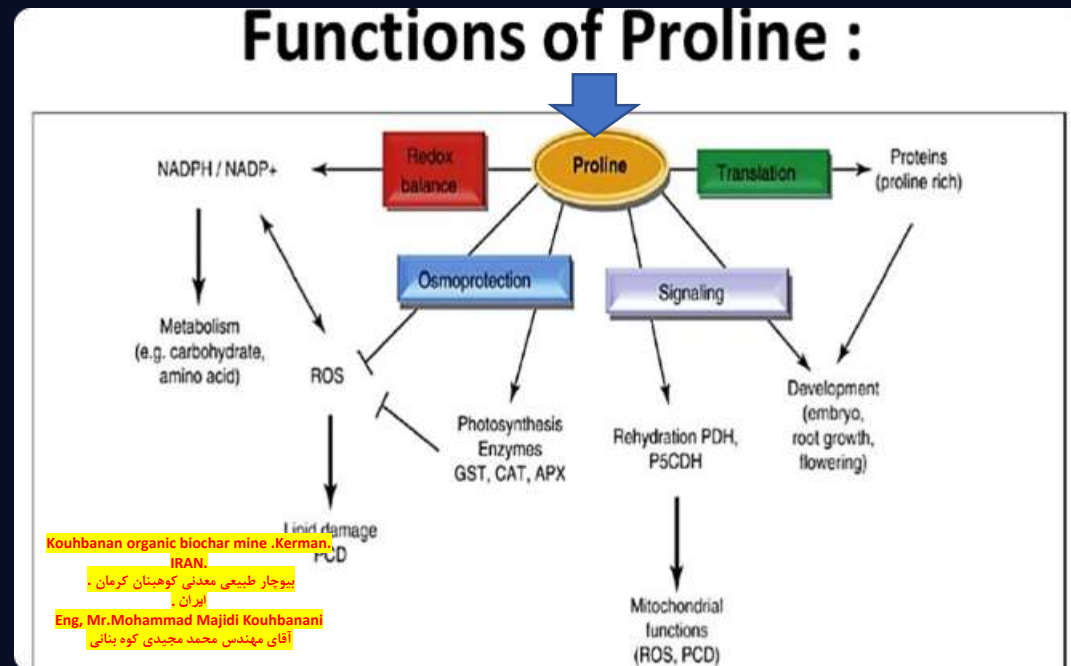
Kerman natural mineral biochar With
Increasing the activity of these enzymes
can play an important role in preventing
frostbite



Increase in phenolic and flavonoid content of leaves as a result of application Kerman natural mineral biochar

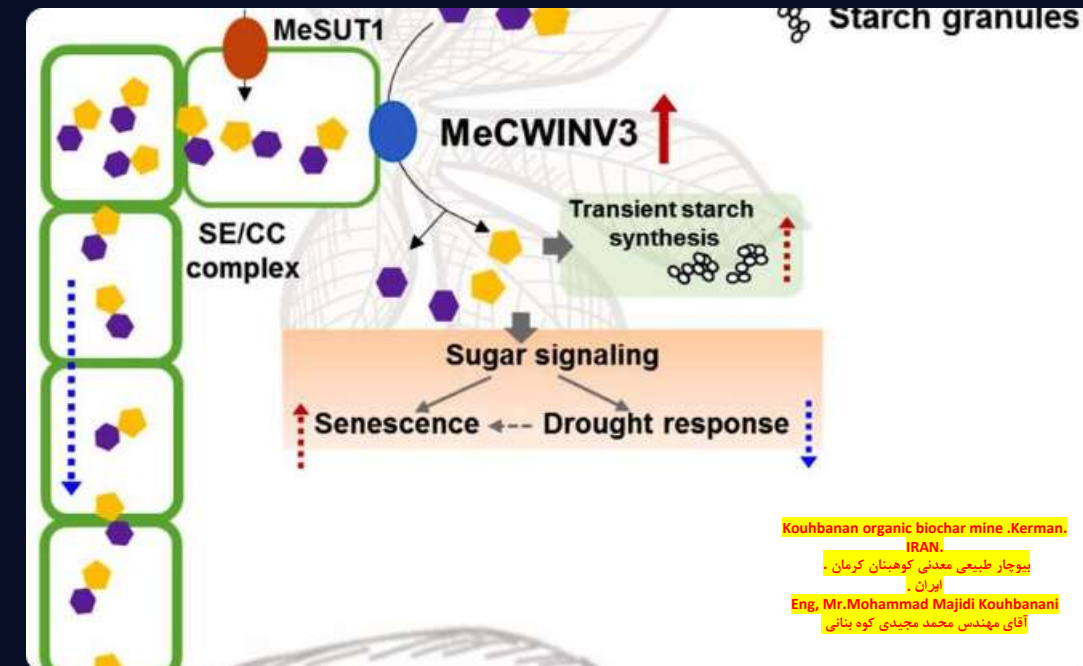


The role of proline and reducing sugars in plant exposure to freezing stress



Proline

Increasing the amount of proline in plants can help in cold stress resistance



Rejuvenating sugars

Increasing the amount of reducing sugars can also be effective in protecting plants against frost

تأثیر بر میزان پرولین



Increase of proline and reducing sugars due to application Kerman natural mineral biochar

the effect Kerman natural mineral biochar

Kerman natural mineral biochar In addition to increasing the amount of proline and reducing sugars, it can be effective in increasing the resistance of plants against cold stress, both proline and soluble sugars can play the role of osmotic protectors when the plant faces cold stress.

Strengthening the plant defense system

Kerman natural mineral biochar By increasing the production of proline and reducing sugars, it can strengthen the health of plants and moderate the dangers caused by cold stress.

تأثیر بر میزان قند احیا

